

5.0 Create Custom Cycle

A “Cycle” is a collection of sounds with a set brightness played over a set duration that when executed simulates the experience of an MRI machine. In cycles, sounds are played concurrently and at the same volume per group. These groups play in succession, providing variation of sounds to simulate the experience of an MRI. The user can create a custom cycle with their choice of sounds and volume per group, as well as overall duration and brightness. Note that a maximum of 15 cycles can be saved to the system at any time.

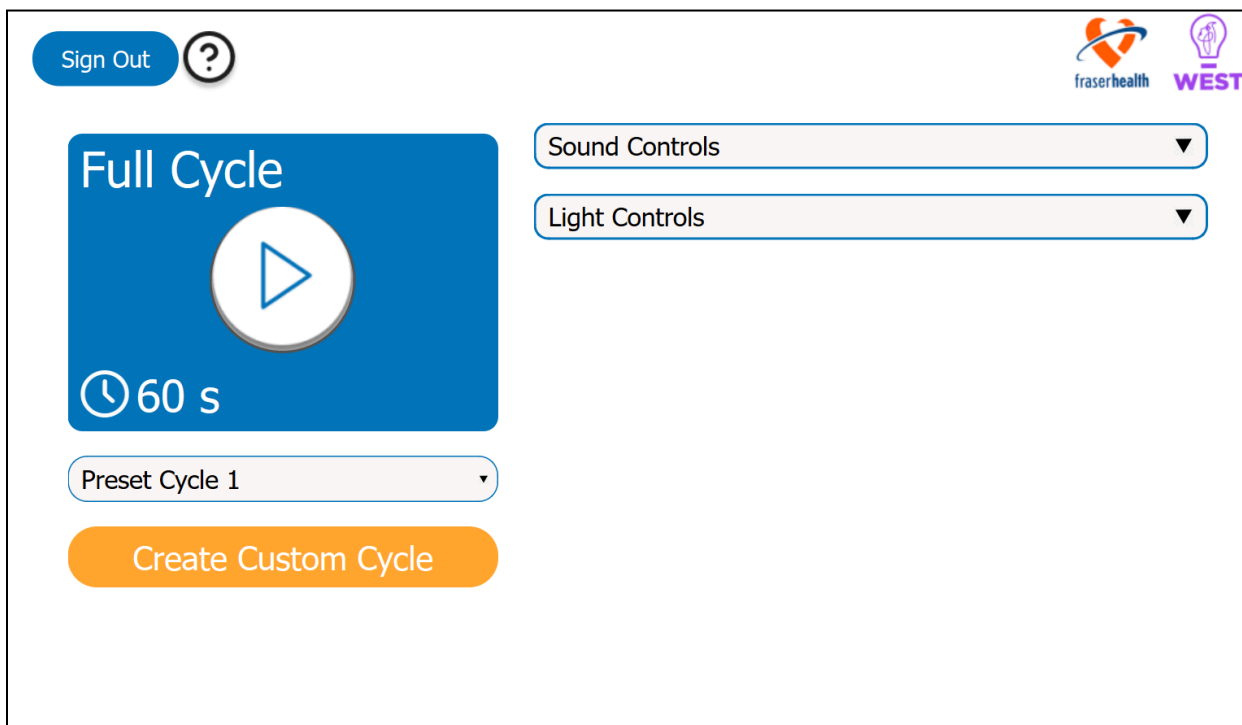


Fig. 5.1 Home Page

Starting on the home page (*figure 5.1*), the user can press the “Create Custom Cycle” button to start creating their own custom cycle. A warning will appear (*figure 5.2*) for safety purposes — make sure any children in the MRI are removed before proceeding as the varying sounds and lights may unsettle them.

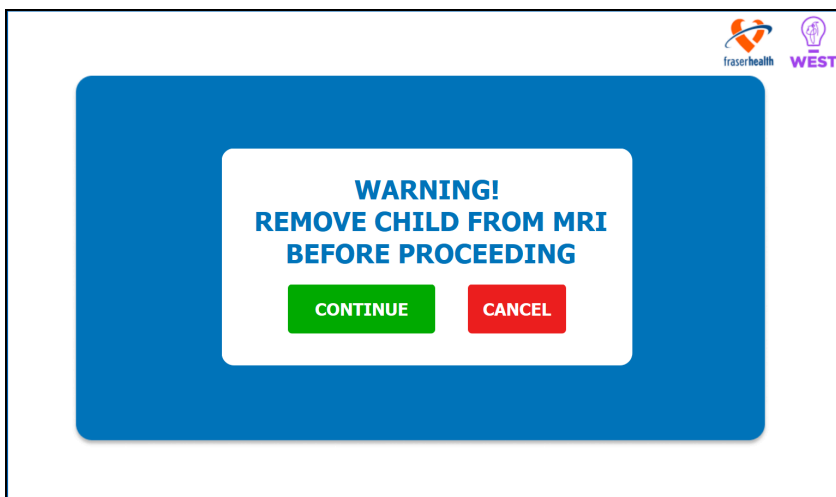


Fig. 5.2 Create Cycle Warning Page

The user will be directed to the cycle duration page (*figure 5.3*), which manages the length of the cycle. The default is set to 5 minutes. The user may increase or decrease the total duration using the plus and minus buttons respectively. Cycles can only be between 1 and 15 minutes in duration.

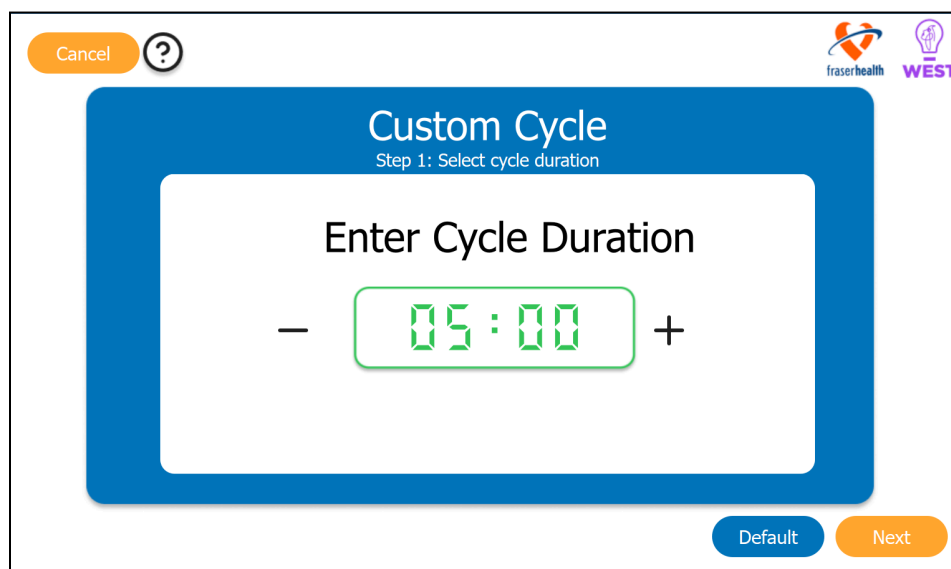


Fig. 5.3 Cycle Duration Selection

After selecting the "Next" button, the user will then be directed to the sound groups per cycle page (*figure 5.4*). Users can modify the group total using the "plus" or "minus" buttons respectively. The default is set to 4 groups, minimum 1 group, maximum 8 groups. As previously mentioned, the total number of groups is the total times the set of sounds playing change throughout the cycle.

Custom Cycle

Step 2: set the number of groups

How many sound groups would you like to have for the new cycle?

— 4 +

Please choose between 1 and 8

Back Default Next

Fig. 5.4 Number of Sound Groups Selection

After selecting “Next”, the user will then be directed to the brightness page (*figure 5.5*), where they can modify the brightness using the “plus” or “minus” buttons respectively. This controls the brightness of the entire cycle.

Custom Cycle

Step 3: set lights

— +

Back Default Next

Fig. 5.5 Brightness Page

After selecting “Next”, the user will be asked to map 1-3 sounds to each group on the sound group mapping page (*figure 5.6*). At least one sound must be selected per group before continuing. If

“Next” is pressed without providing input to a group, a popup message lets the user know that they must complete input before continuing (*figure 5.7*)

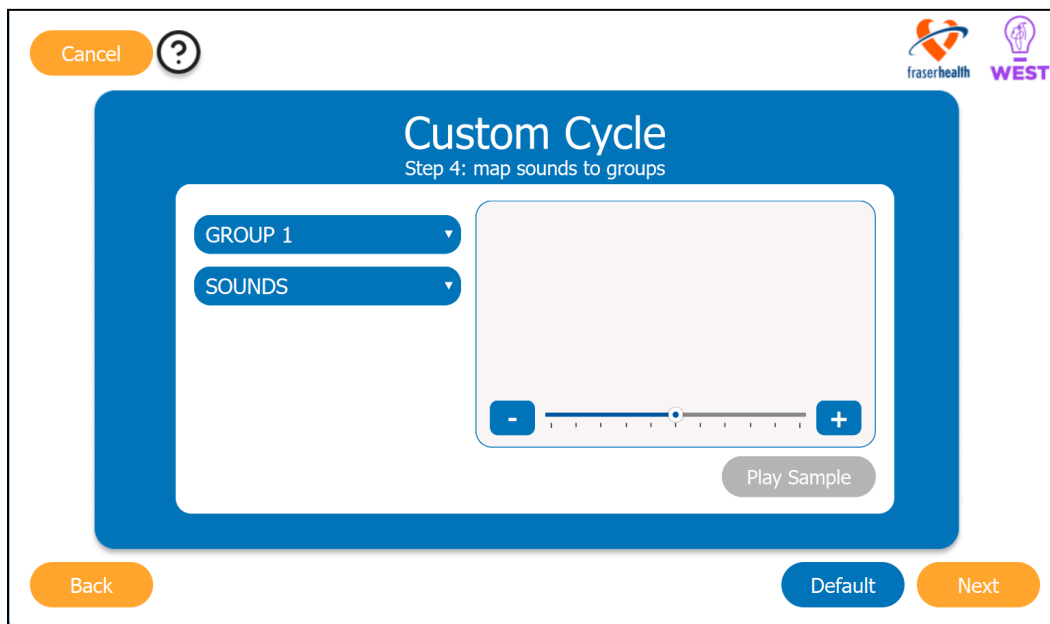


Fig. 5.6 Sound Group Mapping Page

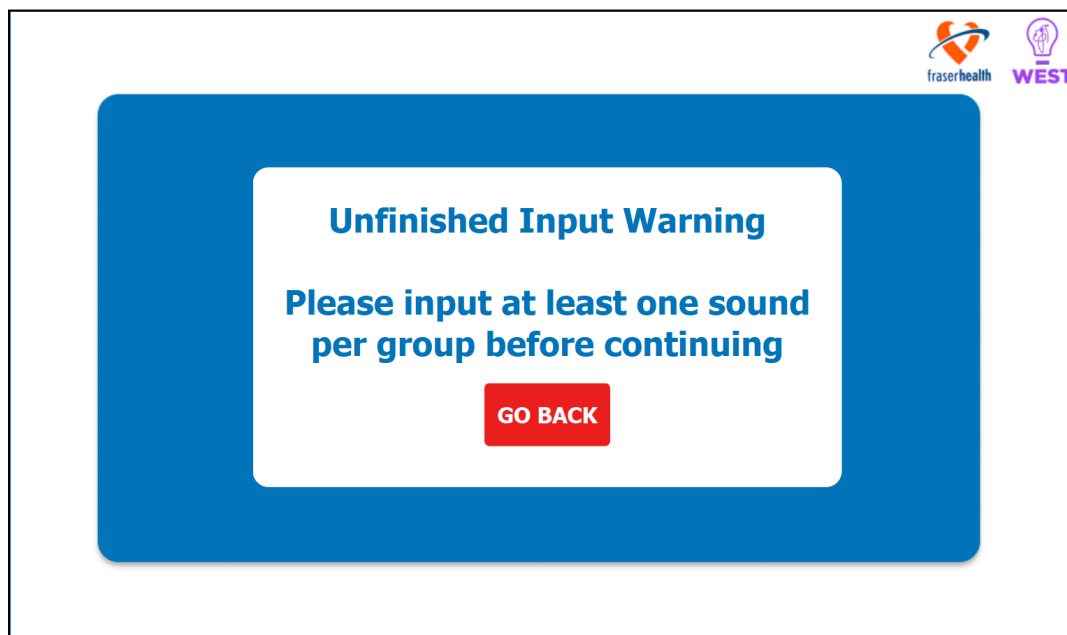


Fig. 5.7 Unfinished Input Warning

By pressing the “GROUP” tab (*figure 5.8*), the tab will expand, allowing the user to select the group they would like to save sounds to. The current group displayed, example “GROUP 1” in *figure 5.8*, is the group currently selected.

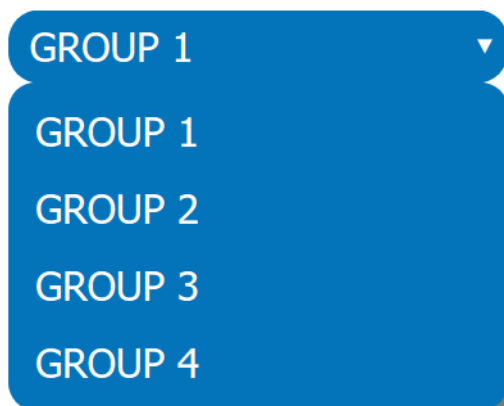


Fig. 5.8 Group Tab Expanded

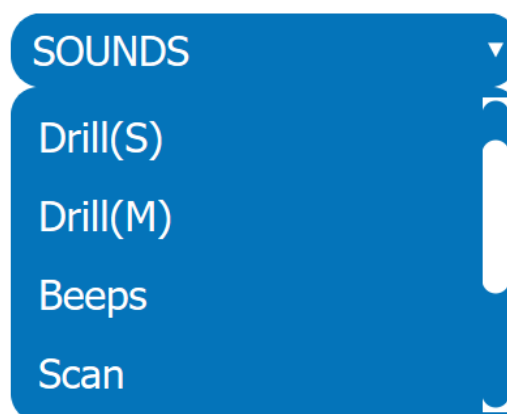


Fig. 5.9 Sound Tab Expanded

Pressing the “SOUNDS” tab (*figure 5.9*) expands the list of selectable sounds to add to a group. Pressing a sound will select it and it will be added to the list of sounds in the sound preview panel found in green on the right of *figure 5.10*. For example, the sounds “Pulse”, and “Drill(S)” have been selected. To remove a sound from the group, simply press the green box containing the sound name, and it will return to the list of selectable sounds. Selecting Default will automatically remove all sounds from the currently selected group.

The volume of a group can be modified using the slider found in the right side of the page. Volume can be increased and decreased by the plus and minus buttons respectively.

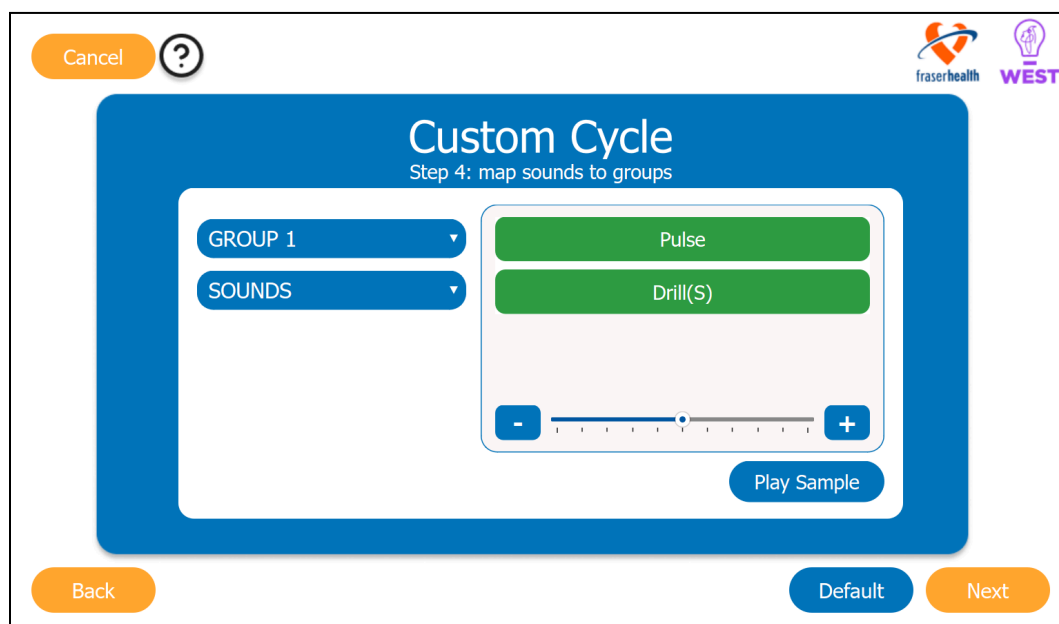


Fig. 5.10 Sound Group Mapping Page With Inputs

Once a sound is added to a group, the user may now play a sample of the sound. Pressing “Play Sample” will play a 5 second sample of the current chosen sound selection. As the sample is playing no further inputs can be added until the sample is finished.

After selecting “Next”, the user will then be directed to the summary page (*figure 5.11*). The blue Cycle Duration, Cycle Lights, or Total Groups buttons show the user’s current selection, and will reroute the user to their respective selection pages to modify as they please.

The sound group preview seen on the right of *figure 5.11* displays the sounds and volume of the selected groups. Other groups can be navigated to via the “<” and “>” arrows respectively. Selecting the blue box holding the name of the current group also reroutes the user to the group selection page

The screenshot shows a mobile application interface for a 'Custom Cycle' summary. The main container is a blue rounded rectangle. At the top, it says 'Custom Cycle' in large white text, followed by 'Step 5: summary' in smaller white text. Inside this container is a white rounded rectangle with a light blue border. On the left side of this white box is a group preview for 'Group 1'. It includes a blue button with 'Group 1' text, flanked by left and right arrow icons. Below this are three input fields: 'Pulse', 'Drill(S)', and 'Volume: 50%'. On the right side of the white box are three blue buttons stacked vertically: 'Cycle Duration: 05:00', 'Cycle Lights: 50%', and 'Total Groups: 1'. At the bottom right of the white box is a green button labeled 'Confirm'. Outside the blue container, at the top left, are an orange 'Cancel' button and a question mark icon. At the bottom left is an orange 'Back' button. In the top right corner of the screen are the Fraser Health and WEST logos.

Fig. 5.11 Custom Cycle Summary Page

Once the user is satisfied with the selection, they may press confirm to finalize the selection. Once a cycle is created it is selected and executable as seen in **Section 4.1: Cycle Selection and Execution**.